

Let $f : [0, \infty] \rightarrow \mathbb{C}$ be a [Measurable Function]. The Laplace transform of f , $\mathcal{L}\{f\}(s)$ or $F(s)$, is defined by

$$\mathcal{L}\{f\}(s) = \int_0^\infty f(t)e^{-st} dt,$$

where $s = \sigma + i\omega$.